

Can Clinical Pharmacists Affect SF-36 Scores in Veterans at High Risk for Medication-Related Problems?

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BACKGROUND. An objective of pharmaceutical care is for pharmacists to improve patients' health-related quality of life (HRQOL) by optimizing medication therapy.

OBJECTIVES. The objective of this study was to determine whether ambulatory care clinical pharmacists could affect HRQOL in veterans who were likely to experience a drug-related problem.

RESEARCH DESIGN. This was a 9-site, randomized, controlled trial involving Veterans Affairs Medical Centers (VAMCs). Patients were eligible if they met ≥ 3 criteria for being at high risk for drug-related problems. Enrolled patients were randomized to either usual medical care or usual medical care plus clinical pharmacist interventions. HRQOL was measured with the SF-36 questionnaire administered at baseline and at 6 and 12 months.

RESULTS. In total, 1,054 patients were enrolled; 523 were randomized to intervention,

and 531 to control. After patient age, site, and chronic disease score were controlled for, the only domain that was significantly different between groups over time was the bodily pain scale, which converged to similar values at the end of the study. Patients' rating of the change in health status in the past 12 months was statistically different between groups, intervention patients declining less (-2.4 units) than control subjects (-6.3 units) ($P < 0.004$). This difference was not considered clinically meaningful. However, a dose-response relationship was observed for general health perceptions ($P = 0.004$), vitality ($P = 0.006$), and change in health over the past year ($P = 0.007$).

CONCLUSIONS. These results suggest that clinical pharmacists had no significant impact on HRQOL as measured by the SF-36 for veterans at high risk for medication-related problems.

Key words: Pharmaceutical care; quality of life; SF-36. (Med Care 2001;39:113–122)

Quality of life (QOL) is a patient-centered outcome measure, and its improvement is espoused as one of the ultimate goals of pharma-

ceutical care.¹ Pharmaceutical care is defined as "... the responsible provision of drug therapy for the purpose of achieving definite outcomes that im-

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prove a patient's quality of life."¹ In contrast to QOL, which refers to an evaluation of all aspects of a person's life, health-related QOL (HRQOL) encompasses and evaluates only those aspects that are dominated or significantly influenced by their personal health.² Assessment of HRQOL among patients within the VA Department is of particular importance. This population is typically poorer, older, and less educated and suffers from more disease than those in the general population.^{3,4}

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Numerous studies have been published that evaluate the impact of pharmacist's services on patients' HRQOL.⁵ The basic premise of these studies is that changes in medication therapy will ultimately lead to better health outcomes and improvements or sustained HRQOL. Most have used the Medical Outcomes Study Short Form 36-item (SF-36) health status questionnaire, a measure of functional status that evaluates 8 domains. Except for 2 studies,^{6,7} no controlled studies have found improvement in HRQOL using the SF-36. Many pharmaceutical care studies have been criticized for poor study design, including small sample sizes, no control group, the pharmacist providing care to both intervention and control patients, lack of randomization, and inconsistent methods of data collection.⁸⁻¹¹ In addition, previous studies provided minimal information on the nature and frequency of the pharmacist intervention. In an effort to overcome these limitations, we designed a study that included those patients who were considered to be most at risk for drug-related problems.¹² The purpose of this study was to examine changes in SF-36 scores for VA patients receiving pharmaceutical care from clinical pharmacists over a 1-year period.

Methods

This was a prospective 9-site (see Acknowledgments for sites), randomized, controlled trial of ambulatory care clinical pharmacists' interventions on economic and humanistic outcomes for VA patients who were at high risk for drug-related problems. The rationale, background, and study design have been previously reported.¹³ This study has been given the acronym name IMPROVE (Impact of Managed Pharmaceutical care on Re-

source utilization and Outcomes in VETERans affairs medical centers). The effect of clinical pharmacists on resource utilization has been reported elsewhere.¹⁴ Institutional Review Board approval was obtained at each participating site and from the Colorado Multiple Institutional Review Board at the University of Colorado Health Sciences Center.

Patient Selection

Patients were considered at high risk for drug-related problems if they met ≥ 3 of the following criteria: (1) were taking ≥ 5 medications, (2) were taking ≥ 12 doses per day, (3) had ≥ 3 chronic medical conditions, (4) had ≥ 4 changes to their drug regimen over the past year, (5) were taking $< 80\%$ of their prescribed chronic scheduled medicines based on pharmacy refill records, or (6) were taking a medication requiring therapeutic drug monitoring.¹² Pharmacy records were used to identify patients meeting the above criteria. The operation and validation of this method to screen for high-risk patients are reported elsewhere.¹⁵⁻¹⁷ Patients meeting ≥ 3 of the criteria were randomized by a central coordinating center to either intervention or control groups. Study sites recruited patients using a list of randomly selected patients sent from the coordinating center.

At each site, medical records (charts and/or computerized data) were reviewed and patients were interviewed to ascertain whether they met additional inclusion/exclusion criteria. Patients were included in this study if they (1) were a VA Medical Center (VAMC) patient for ≥ 12 months, (2) anticipated receiving medical care from their participating VAMC for the duration of the study, and (3) lived within reasonable proximity to allow frequent visits (possibly 1 per month) and had transportation to and from the VAMC. Patients were excluded from this study if they (1) had been seen in a pharmacist-managed clinic within the previous 12 months; (2) had a medical condition, such as terminal cancer, that limited their life expectancy to ≤ 12 months; (3) required services from mental health or a similar clinic; (4) had a poor understanding of written and spoken English; or (5) were visually impaired (not being able to read standard newsprint). Patients who had previous contacts with clinical pharmacists were excluded because it was thought that these patients would have had some type of intervention.

Patients attending mental health clinics were excluded because general practice clinical pharmacists would probably have minimal impact on these patients. The requirements of speaking and understanding English and the ability to read newsprint were to ensure that patients could complete self-administered satisfaction questionnaires and the health status survey.

All patients signed informed consent and were not compensated for participation. When patients were enrolled in the study, clinical pharmacists who were blinded to treatment group conducted a review of medical records to document medical conditions and medications.

Pharmacist Intervention

Patients enrolled in the intervention group were seen by clinical pharmacists practicing in ambulatory clinics. A total of 78 pharmacists provided care to intervention patients in this study. The study protocol indicated that each patient should have a minimum of 3 visits with the clinical pharmacist over the course of the study, but patients could be seen as frequently as deemed necessary to ensure appropriate care. Control patients were not seen by clinical pharmacists in this study. To prevent contamination by clinical pharmacists, for control patients, a variety of measures were used (eg, marking medical records with patients' group assignment, adding a note to the electronic medical record, providing printed lists of enrolled patients to pharmacists). Despite these measures, 1 control patient was inadvertently intervened on by a clinical pharmacist. On discovery of this error, the patient was disenrolled from the study and not included in the results.

Intervention visits with clinical pharmacists were to occur between or concurrent with patient visits with their primary care provider or other health care providers. This study did not target patients with specific medical conditions, so the disease states evaluated by pharmacists were varied. Each clinical pharmacist was to practice according to his/her defined scope of practice within his/her respective VAMC. In all sites, clinical pharmacists worked in conjunction with physicians and other health care providers to provide care to patients. Clinical pharmacists providing care in this study were not dispensing pharmacists. These pharmacists worked proactively to meet the needs of the patient. Medical records were reviewed and

patients were interviewed to optimize medication therapy. Specific activities performed by these pharmacists included assessing appropriateness of medication therapy; physical assessments, such as measuring blood pressure; establishing goals of medication therapy; assessing medication compliance on a consistent basis via computer-generated medication refill profiles; conducting in-office laboratory tests, such as blood glucose tests via finger sticks; identifying nontreated disease states that may benefit from pharmacological therapy; checking medication profiles for drug-drug, drug-food, and drug-disease interactions; monitoring medications for therapeutic effect and toxicity by requesting laboratory tests; providing patient education; and referring patients to primary care physicians, specialists, and other health care providers. Depending on the site and a pharmacist's scope of practice, medication therapy could be started, stopped, or changed.

A standardized form was used by clinical pharmacists to document those activities performed when contacting a patient. This form categorized the amount of time spent with patients into <5 minutes, 5 to 14 minutes, 15 to 29 minutes, and ≥ 30 minutes. Pharmacists also indicated whether the contact occurred in person or via telephone. Each disease or condition addressed was recorded. Common medication-related problems were listed to allow pharmacists to indicate the type of pharmaceutical care intervention provided.

HRQOL Assessment

HRQOL was assessed via the SF-36. The SF-36 was chosen because of its applicability to a variety of disease states, brevity, and ease of administration and for comparison with national norms. In addition, it has been found to be reliable and valid in general populations, ambulatory populations, and populations with chronic diseases.¹⁸⁻²¹ The SF-36 consists of 8 separate domains: physical functioning, general health perceptions, role physical, role emotional, bodily pain, social functioning, mental health index, and vitality. In addition, 1 question assesses change in health status in the past year. Each domain is transformed to a 0 to 100 scale, with higher scores representing higher HRQOL.

The SF-36 questionnaire was administered to patients in both groups at enrollment, as well as at 6 ± 1 and 12 ± 1 months. Surveys were self-

administered to study patients in a quiet area (eg, exam room, research office) and then transferred to 2-part forms by research staff. If the patient was unable to visit the medical facility to complete the SF-36, the questionnaires were mailed to participants. Each study site was visited by a member of the IMPROVE research team to verify accuracy of transcribing for all completed questionnaires. Survey responses were subsequently key punched with double-entry verification. For those domain scales in which the patient did not answer 1 question, the mean value from the other domain items answered by that patient was imputed. In situations in which the patient missed >2 questions or the domain had <3 items, the domain score was not calculated.

Sample Size and Statistical Analysis

The study was powered to detect a clinically meaningful difference in SF-36 scores. A difference of 5 points in changed SF-36 domain scores was deemed clinically meaningful.²² Detecting a 5-point difference on the 8 domains between 2 experimental groups using repeated-measures analysis would require a per-group sample ranging from 132 for the mental health index scale to 468 for the role physical scale.²² These estimates were calculated on the basis of the assumption of a 2-tailed test for which the α error is 0.05 and power is 0.80, and the intertemporal correlation between scores was 0.60. Another desired end point for this study was to detect differences in health care costs between treatment groups. However, no data were available to determine the required sample size for economic end points. Therefore, the sample size of a 1,000 evaluable patients was set on the basis of financial and logistical resources under the assumption that this sample size was sufficient to demonstrate differences in HRQOL.

Data were analyzed with an intent-to-treat analysis. Multivariate repeated-measures analysis of variance was used to compare scores for each SF-36 domain. Independent variables in the analysis included site, group (control or intervention), group $\times$ site interaction term, time period (baseline or 6 and 12 months), site $\times$ time period interaction term, group $\times$ time period interaction, group $\times$ site $\times$ time period interaction term, patient age at study enrollment, and chronic disease score (CDS).²³ The primary analyses were conducted using data from only

those patients with complete data across all 3 time points. This may lead to biased results if there are unequal dropout rates.²⁴ To assess the extent of dropout bias, comparisons were made between those patients who disenrolled and those remaining in the study.

To account for multiple comparisons, the α error was set to $P < 0.01$. Data were analyzed with SAS version 8.0. Repeated-measures analysis of variance was conducted using the PROC MIXED function within SAS with the assumption of compound symmetry for the within-subjects variance-covariance matrix.

Results

Study Enrollment

A total of 1,054 patients were enrolled, 523 randomized to the intervention group and 531 to control. A total of 950 patients completed 6-month follow-up questionnaires, and 931 subjects completed the study. The maximum number of patients enrolled at any 1 site was 200; the minimum number was 42. Five locations enrolled ≥ 100 patients in the study. Only 1 site had <65 patients. Intervention or control group enrollment was balanced in all sites. More subjects in the control group completed the study (483, 91.0%) compared with the intervention group (448, 85.7%); this difference was statistically significant ($P = 0.007$). There were 21 and 17 deaths in the intervention and control groups, respectively.

Baseline Characteristics

Table 1 displays the baseline characteristics of the 2 treatment groups. The 2 groups were similar in demographics and medical conditions at study enrollment. Hypertension was present in more than two thirds of patients. Angina, hyperlipidemia, and arthritis was present in 46%, 41%, and 36% of the patients, respectively. SF-36 scores at baseline also are shown in Table 1. The scores for the intervention group were lower for all domains compared with the control group, but differences were not statistically significant at the $P = 0.01$ level.

Clinical Pharmacist Interventions

During the course of the 12-month period, clinical pharmacists recorded a total of 1,855 con-

TABLE 1. Baseline Characteristics

Item	Intervention (n = 523)	Control (n = 531)	P
Gender, n (%)			
Male	502 (96.4)	511 (96.2)	0.83
Female	21 (0.04)	20 (0.04)	
Marital status, n (%)			
Single	40 (7.6)	44 (8.3)	0.94
Married	358 (68.5)	360 (67.8)	
Divorced	80 (15.3)	85 (16.0)	
Widowed	45 (8.6)	42 (7.9)	
Age, y	66.8 ± 10.2	66.6 ± 10.0	0.77
Select chronic conditions, n (%)			
Hypertension	358 (68.5)	353 (66.5)	0.49
Angina	241 (46.1)	248 (46.7)	0.83
Hyperlipidemia	208 (39.8)	229 (43.1)	0.27
Arthritis	177 (33.8)	203 (38.2)	0.14
Diabetes	177 (33.8)	158 (30.0)	0.15
GERD	153 (29.2)	127 (23.9)	0.05
COPD	109 (20.8)	119 (22.4)	0.54
Baseline SF-36 scores			
Physical functioning index	49.9 ± 27.6	50.3 ± 27.4	0.363
Role physical	36.4 ± 41.0	39.7 ± 42.3	0.182
Bodily pain	46.6 ± 25.6	50.0 ± 25.0	0.026
General health perceptions	49.1 ± 22.0	48.6 ± 22.5	0.749
Vitality	45.1 ± 21.2	45.9 ± 21.9	0.478
Social functioning	65.2 ± 28.4	67.7 ± 26.3	0.133
Role emotional	58.0 ± 42.6	62.7 ± 43.4	0.067
Mental health	69.6 ± 19.9	72.0 ± 17.9	0.033
Change in health	52.2 ± 25.9	52.6 ± 25.2	0.802

GERD indicates gastroesophageal reflux disease; COPD, chronic obstructive pulmonary disease.

tacts with the 523 intervention patients. The mean number of contacts was 3.5±2.3 (range, 0 to 18). Only 27.9% of patients had ≤2 contacts; 37 intervention-assigned patients had no contacts with pharmacists. More than 73% of the patient contacts with a clinical pharmacist were ≥15 minutes. A more complete description of the pharmacists’ interventions is reported elsewhere.²⁵

Effect of the Intervention

Changes from baseline to 6 and 12 months for SF-36 scores are shown in Table 2. There was a general decline in SF-36 scores across all the domains except the mental health index scale. During the 12-month study, SF-36 scores signifi-

cantly declined for the physical functioning, role physical, bodily pain, general health perceptions, vitality, and social functioning domains ($P \leq 0.001$). Respondents’ rating of the change in their health over the last year also significantly declined. In all of the multivariate models, site was a significant predictor of SF-36 score ($P \leq 0.0001$), but the site×group, site×time period, and site×group×time period interaction terms were not statistically significant. Examination of SF-36 domain scores across sites found that 1 site had consistently higher scores for all of the scales. In addition, the CDS was a statistically significant predictor of SF-36 domain scores ($P \leq 0.003$) for all domains except for change in health over the past year ($P = 0.0243$). Age was a statistically signifi-

TABLE 2. Change From Baseline in SF-36 Scores

	Baseline to 6 Months			Baseline to 12 Months			<i>P</i> *
	Intervention, Mean (SE)	Control, Mean (SE)	Difference Between Groups	Intervention, Mean (SE)	Control, Mean (SE)	Difference Between Groups	
Physical functioning	−4.9 (0.97)	−3.4 (0.88)	−1.5	−5.3 (0.97)	−6.1 (0.97)	0.8	0.412
Role physical	−3.5 (1.80)	−4.3 (2.06)	0.8	−4.3 (1.98)	−8.2 (2.00)	3.9	0.245
Bodily pain	−0.8 (0.96)	−3.3 (0.91)	2.5	−0.3 (1.04)	−4.8 (1.04)	4.5	0.004
General health perceptions	−1.6 (0.76)	−2.2 (0.73)	0.6	−2.4 (0.81)	−5.3 (0.76)	2.9	0.026
Vitality	−0.4 (0.78)	−2.2 (0.84)	1.8	−1.3 (0.88)	−5.0 (0.89)	3.7	0.019
Social functioning	−1.0 (1.12)	−3.7 (1.20)	2.7	−3.4 (1.29)	−5.0 (1.26)	1.6	0.302
Role emotional	−2.6 (2.15)	−3.4 (1.94)	0.8	−0.3 (2.28)	−7.4 (2.31)	7.1	0.065
Mental health	−0.5 (0.80)	−1.4 (0.73)	0.9	0.1 (0.80)	−2.3 (0.77)	2.2	0.029
Change in health	−1.1 (1.28)	−4.8 (1.31)	3.7	−2.4 (1.52)	−6.3 (1.27)	3.9	0.004

*Represents differences between groups over time controlling for age, CDS, and site.

cant predictor for the bodily pain (*P* = 0.003) and mental health index (*P* = 0.010) domains.

The only SF-36 domain with significant differences between groups over time was the bodily pain domain (*P* = 0.004). This scale converged over time to similar values at the end of the study. The general health perceptions domain declined on average 2.4 units in the intervention group compared with 5.3 units for the control group, but this difference was not statistically significant when controlling for age, site, and CDS (*P* = 0.026). The largest change in average score between groups was for the role emotional domain, for which the intervention scores increased by 0.3 units but the control group value declined on average by 7.4 units, but this difference was not statistically significant (*P* = 0.065).

As mentioned previously, the dropout rate for the intervention group (12.5%) was greater than for the control group (9.0%). Analysis of baseline SF-36 scores for patients with complete data compared with those who discontinued the study found no significant differences across all the SF-36 domains, except for the mental health index, where subjects who completed the study had higher scores than those who did not (71.5±18.4 versus 66.0±21.7, *P* = 0.006).

Dose Response of the Intervention

To determine whether there was a dose response, linear regression analysis was conducted with the change in each domain score from baseline to 12 months used as the dependent variable

and the number of contacts with clinical pharmacists, patient age at study enrollment, and CDS used as independent variables. For every additional pharmacist contact, domain scores for the general health perceptions, vitality, and change in health over the past year improved by 0.65 (*P* = 0.004), 0.70 (*P* = 0.006), and 1.1 (*P* = 0.007) units, respectively. Even though none of the other domain scores were significantly related to the number of clinical pharmacist contacts, the regression coefficients were all in the same direction, toward improving HRQOL.

Discussion

This study found that clinical pharmacist interventions had no meaningful effect on HRQOL for veterans at high risk for drug-related problems, but there was evidence to suggest a dose-response relationship. Most SF-36 domain scores were declining over time, which can be attributed to the sample of mostly elderly men with significant comorbidity, but little difference between groups was observed. The intervention group declined 1.7 units on the general health perceptions scale compared with 5.1 units for the control group. This difference of 3.4 units, on a scale that ranged from 0 to 100, was considered not clinically relevant. The only scale for which the difference in mean change was >5 units was the role emotional domain (7.1 units). The only component of the SF-36 for which intervention patients declined significantly less than control patients was the question concerning change in health over the

past year ($P = 0.004$). A statistically significant dose-response relationship was observed for the number of clinical pharmacist contacts. However, it would take 7 interventions to yield a 5-unit change for the general health perceptions domain.

Pharmacists' inability to influence patients' HRQOL in this study is consistent with other controlled studies examining pharmacist interventions.^{26–28} One of these studies was conducted in patients with hypertension²⁷ and another in persons with diabetes.²⁷ Hanlon et al²⁶ found no differences in SF-36 among an elderly veteran population that received a pharmacist intervention. In another study of hypertensive patients, Carter and colleagues⁶ found significant differences on the physical functioning and role physical domains of the SF-36. Cody and colleagues⁷ used a mailed SF-36 survey to estimate HRQOL for a subsample of the study population in the Kaiser Permanente/USC Patient Consultation Study. The Kaiser study found changes of <5 units over a 2-year period for all domains of the SF-36. Varma et al²⁹ reported the results of a randomized, controlled pharmaceutical care intervention for patients with CHF who completed both the SF-36 and the Minnesota Living With Heart Failure Questionnaire. Significant differences were observed between intervention and control patients for the physical functioning and mental health domains, but the analysis failed to take into account baseline differences between the groups. Other studies have found significant differences in HRQOL using other instruments (eg, Health Status Questionnaire 2.0, disease-specific questionnaires), but the study designs were often not optimal^{30–33} or the sample size was too small.^{30–34} The inconsistent ability for pharmacists to have an effect on HRQOL has caused some researchers⁵ to question whether this was the intended effect of pharmaceutical care as defined by Hepler and Strand.¹

Pharmacists are not alone with respect to not having a dramatic impact on improving HRQOL. A study involving a nurse-coordinated education, compliance, and access to care program found no differences in patients' SF-36 scores compared with usual medical care.³⁵ Another study also found no significant difference in SF-36 scores among frail elderly subjects randomized to a study of nurse case management compared with usual medical care.³⁶ A randomized study of a senior self-management and disability prevention program led by a geriatric nurse practitioner found no

differences in SF-36 scores over 1 year.³⁷ One study has demonstrated a difference in SF-36 scores among specialists compared with generalists for persons with asthma,³⁸ but this study was cross sectional, and patients who saw allergists had more severe asthma and had greater opportunity to improve.

There are several plausible factors that may have affected the results of this study. The primary analyses were conducted using data from only those patients with complete data across all 3 time points. There was a small but statistically significant greater dropout rate for the intervention group that might have led to biased results. There are several methods to account for missing data. The first method is to carry forward the last observation for each subject with missing values. The second method is to impute the average score of patients in that arm of the study at a specific time (6 or 12 months). The last method is to impute the average score for all patients across both study arms at 6 and 12 months, respectively. All of these methods of imputation were performed, and the results yielded results nearly identical to those subjects who completed the SF-36 at the 3 required observation points.

Another potential problem with nonsignificant results is whether the study had sufficient power to detect a difference. Post hoc power analyses were conducted to determine the power of the study to detect a 5-unit difference between groups using methods specified by Cohen,³⁹ with an α error rate of 0.01. Power estimates were calculated for each of the domains: physical functioning (99%), role physical (33%), vitality (97%), general health perceptions (99%), bodily pain (89%), role emotional (23%), social functioning (71%), mental health index (99%), and change in health over the past year (72%). Therefore, it appears that this study had adequate power to detect a 5-unit difference for 5 of the 8 domains. A criticism of the SF-36 power tables is that they assume "normal" distributions of the transformed data. Julious et al⁴⁰ suggested that the SF-36 is nonparametric and that proportional odds must be used to calculate sample size. If a 1-value (from a 5 to a 4) change is the desired minimal important difference, then sample sizes range from 64 for the role emotional to 1,401 to the bodily pain domain.⁴⁰ Under the assumption of nonparametric data, the IMPROVE study has sufficient subjects in each arm to detect a negative 1-value change for 5 of the 8 domains.

The SF-36 may not be sensitive enough to detect changes in health status that are attributable to pharmacist interventions. Considered to be a generic HRQOL instrument, questions asked by the SF-36 are not disease or provider specific. Substantial numbers of patients in this study had conditions such as hypertension or high cholesterol that are asymptomatic until advance stages of disease. Several studies have documented that pharmacists can affect blood pressure control,^{6,27} hemoglobin A_{1c} levels,²⁸ and heart failure.⁴¹ Improving control of these diseases can result in clinically significant outcomes that may not be perceived by the patient using generic HRQOL instruments. Another limitation of the study was that patients were studied for only 1 year. There was a slight trend favoring intervention patients, and perhaps a longer study might yield larger changes, especially because there was a slight dose-response relationship. It also is possible that IMPROVE patients were "too sick" to improve HRQOL or to prevent further decline in HRQOL, a floor effect. Across the domains, SF-36 scores were relatively low compared with national norms.²² It is possible that a floor effect influenced the results of this study, whereby patient scores could decline very little because they were already low.

Conclusions

The findings of this large controlled trial lead us to question the utility of using generic HRQOL instruments, such as the SF-36, to measure the impact of clinical pharmacist services on HRQOL. Most studies of pharmaceutical care are limited in terms of sample size and will lack statistical power to detect significant differences using the SF-36.^{9,10} Future evaluations of pharmaceutical care should focus on clinical measures of disease or use disease-specific HRQOL instruments. The number of disease-specific instruments is expanding rapidly, and these measures may be more sensitive to interventions provided by pharmacists. In addition, work by Murawski and Bentley⁴² on a pharmaceutical therapy-related quality of life instrument may lead to a useful tool for measuring the impact of pharmacists.

The provision of pharmaceutical care by clinical pharmacists appears to have no effect on HRQOL as measured by the SF-36 in veterans at high risk for drug-related problems. A dose-response rela-

tionship was observed, but a 5-unit change in general health perception would require 7 pharmacist interventions. Investigators examining the impact of pharmaceutical care or similar pharmacist interventions should focus their research efforts using disease-specific HRQOL measures that may be more sensitive to change.

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